

Name: _____



Checking a claim

Task A: Analysing headlines

1) A magazine article titled “Look at the savings you would make if tax was abolished” states the figures below.

Correct the prices of each item without VAT.

Television costing £450	Would cost only £360	Car seat costing £180	Would cost only £171
(without 20% VAT)		(without 5% VAT)	

2) In a raffle there is a 1 in 2180 chance of winning a prize. A headline reads “Just buy 2 tickets and increase your chance of winning by 100%”.

- What is the chance of winning with one ticket, as a percentage?
- What is the chance of winning with two tickets, as a percentage?
- What is the percentage point difference in the percentage chance of winning between buying 1 and 2 tickets?
- Explain why, although technically correct, the claim could be misleading.
- Use trial and improvement to find how many tickets you would need to buy to increase the probability of winning to 1%.

3) Explain the mistake in each headline.

- a) “Local employment rates fell by 10% last year and 15% the previous year. In just 2 years, employment rates have fallen by 25%.”

- b) “28% of men and 22% of women say they only clean their teeth once a day. A whopping 50% of adults are not cleaning their teeth twice a day.”

- c) “Decrease your food expenditure by 300% a week by taking your own lunch to work (based on a takeaway lunch cost of £28 a week and home made lunch cost of £7 a week).”

Task B: Estimating to check

1) An article reads: “The average person will spend 100 000 hours on their phone in their lifetime.”

a) Estimate how long you think the average person will spend on their phone in a lifetime. Write down any estimations you make.

b) Does the article’s claim sound reasonable?

2) A newspaper states that “5% of the UK has a car parked on it.”

a) Estimate how many households there are in the UK.

b) Estimate how many cars there are in the UK.

c) Estimate how much space is taken up by these cars.

d) Assume the area of the UK can be modelled by the two rectangles shown. What is the approximate area of the UK?

e) Does the newspaper’s claim seem valid?

